BridgeFinder: Discovering Cross-Community Research Collaborators at UIUC

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Abstract

Students seeking research mentors often know an instructor but lack a route into another research community. BridgeFinder uses a weighted UIUC co-authorship graph to recommend researchers outside a seed instructor's community and explain possible introductions. The challenge is balancing topical relevance with network diversity while avoiding claims that a missing co-authorship proves a social gap. We have built an OpenAlex ingestion pipeline, two real-data corpora, a three-screen web prototype, temporal backtests, a citation analysis, and a formative study with five students. The deterministic temporal backtest tests the same ranker used in the interface against popularity, centrality, topic-only, and Adamic–Adar baselines; its results show modest gains rather than satisfaction of all improvement targets. A citation comparison finds a mean ratio of 1.053 ($p = 0.0601$), providing no significant evidence of a cross-community advantage. Feedback led to an instructor-anchor entry point, an explicit missing-tie legend, and prominent introduction paths. Shared model code aligns the interface and evaluation, while a browser-validated study workflow supports the next comparative evaluation.

1 Problem, objective, and contribution

A student who enjoyed a computing course may want to apply those skills in biology or agriculture but know neither whom to approach nor who could introduce them. Keyword search supplies names; an existing co-author network supplies routes, but tends to keep exploration close to familiar groups. Our objective is to help UIUC undergraduate and early-stage graduate students find a relevant researcher outside a familiar instructor's collaboration community and understand a plausible next step toward contact.

The problem is challenging for three reasons. Students may have no publications from which to build a personal profile. Topical similarity alone can return familiar researchers, while network distance alone can return irrelevant ones. Finally, published co-authorship is an incomplete proxy for acquaintance, trust, and willingness to mentor. BridgeFinder therefore combines topical overlap with community boundaries and brokerage features, while treating paths as suggestions to verify rather than guaranteed introductions.

Our contributions at midterm are (1) a weighted graph pipeline with a temporally separated evaluation on two UIUC datasets; (2) a working seed–map–recommendation interface; (3) qualitative feedback linked to concrete design changes; and (4) preliminary results and a comparative evaluation plan. These establish feasibility, not improved mentoring outcomes or causal effects of brokerage.

2 Related work and design implications

Structural holes and weak ties. Granovetter explains how weak ties connect otherwise separated social circles [6]; Burt relates brokerage to access to nonredundant information and ideas [2, 3]. These theories motivate exploration beyond a student's immediate group. Their strength is explaining why diverse contacts can matter; their limitation here is that co-authorship data does not measure tie strength or prove that a suggested edge creates useful brokerage. We operationalize community boundaries as a testable proxy rather than claiming a direct measurement of social opportunity.

Academic discovery tools. Illinois Experts [11] and Google Scholar provide useful profile and publication search. Connected Papers and ResearchRabbit support visual literature exploration through relationships among papers. Their strength is fast discovery of expertise or related literature; their limitation for our task is that their central workflow does not explain a student-to-faculty introduction chain. BridgeFinder adapts visual graph exploration to people and makes an intermediate contact explicit. We do not claim that these systems lack all relational features.

Collaboration prediction. Liben-Nowell and Kleinberg [7] show how local network measures predict future links. Adamic–Adar rewards shared neighbors and is an interpretable, inexpensive baseline, but can favor familiar neighborhoods. Tang et al. [10] address cross-domain collaboration recommendation, demonstrating that topical and relational evidence can be combined. Its strength is making cross-domain matching explicit; its task and model do not directly establish whether a student can act

on a displayed introduction. We compare against topic-only and local-topology models, and evaluate actionability separately.

Organizational context. Dahlander and McFarland [4] study tie formation and persistence in a research organization. The longitudinal evidence motivates treating institutional context and intellectual similarity as important. Its limitation for our design is that an observational explanation is not itself an interactive discovery workflow. Our interface exposes candidate explanations; the evaluation must still test whether students find them useful.

3 Course concepts and network analysis

3.1 Graph construction and temporal scope

We use OpenAlex [9] works dated 2015–2024 associated with UIUC's ROR identifier `047426m28`. Nodes are author identifiers with UIUC affiliation on harvested works. Each retained paper contains at least two UIUC-affiliated authors; external collaborators are omitted. The 5,000-work harvest yields 2,493 such papers and 5,396 author records; the 25,000-work harvest yields 13,229 papers and 16,604 author records. These capped harvests are samples, not a campus census.

For retained UIUC authors A_p on paper p, the undirected edge weight is

$$w_{ij} = \sum_{p:\, i,j \in A_p} \frac{1}{|A_p| - 1}.$$

Fractional weighting limits the contribution of each large team without discarding it. The backtest includes large teams (up to 70 and 92 retained authors in the two corpora); the separate citation script filters to at most 25 retained authors. Thus the two analyses do not use identical graph construction. Training uses 2015–2019 papers, evaluation uses 2020–2024 papers, and backtest topics are each author's ten most frequent training-paper topics. Authors with no training edges are outside the prediction population.

3.2 Communities, brokerage, and routing

We use weighted Louvain modularity optimization [1] to identify collaboration communities, rather than assuming that departments are communities. Modularity compares within-group edge weight with a strength-preserving null model:

$$Q = \frac{1}{2W} \sum_{ij} \left(w_{ij} - \frac{s_i s_j}{2W} \right) \mathbf{1}[c_i = c_j].$$

Here $s_i = \sum_j w_{ij}$ and $W = \sum_{i<j} w_{ij}$. Girvan–Newman [5], an alternative covered in community discovery, repeatedly removes high-betweenness edges and offers an interpretable hierarchy, but repeated shortest-path computations are expensive. Louvain is a practical choice for repeated analysis of thousands of authors; its resolution limit and stochastic partitions remain limitations. We do not treat its output as verified disciplinary labels.

The pilot training graph contains 3,642 nodes and 16,408 edges; the scaled graph contains 9,042 nodes and 36,748 edges. The backtest partitions the union of all nontrivial connected components, not just the largest component. The canonical, seeded run finds 344 pilot communities with $Q \approx 0.949$. This high modularity indicates strong separation of observed collaboration groups, but disconnected components can inflate it. It does not establish that all groups are socially isolated or that crossing them is useful. The graph structure motivates both cross-community filtering and an explicit distinction between reachable and unreachable candidates.

For candidate v, Burt autonomy is $H(v) = \max(0, 1 - C_v)$, with

$$C_v = \sum_{j \in N(v)} \left(p_{vj} + \sum_{q \in N(v)} p_{vq} p_{qj} \right)^2, \quad p_{ij} = \frac{w_{ij}}{\sum_k w_{ik}}.$$

The implementation assigns $H = 0$ to degree-zero/one nodes. This feature measures the candidate's existing neighborhood, not the marginal benefit of a new edge. Boundary spanning toward seed u is $B(v, u) = |N(v) \cap C(u)|/|N(v)|$, where $C(u)$ now denotes the set of nodes in u's community.

Candidates are non-neighbors in another community. A distance-two candidate has a shared co-author who could be asked about an introduction. Distance three implies two intermediaries; no discovered short path calls for cold outreach. The student-to-seed anchor is user supplied and is not an observed graph edge. These ideas connect structural holes, triadic closure, community discovery, and small-world navigation from the course. Our computation is centralized graph analysis, not an implementation of decentralized local search, and graph distance does not measure sociological tie strength.

3.3 Offline score and ablation

Let S be Jaccard overlap of training-topic sets. The backtest ranks candidates by

$$\Phi(u, v) = 0.60S + 0.12H + 0.12B + 0.16P. \qquad (1)$$

$P = 1$ for two-hop candidates, 0.5 for candidates with a simple three-edge path and no two-hop connection, and 0.1 otherwise. The final case includes longer paths and disconnected components; it is not a reachability probability. These fixed heuristic weights prioritize relevance. A Topic+Path ablation uses $S + 0.5c_3/(1 + c_3)$, where c_3 is the number of simple three-edge paths. Comparing the ablation with the full model tests whether autonomy and boundary spanning add value beyond topic and route information.

4 Design and implementation

4.1 A demonstrable core flow

The implementation separates expensive graph preparation from browser interaction. `fetch_openalex.py` produces processed datasets; `backtest.py` and `citation_impact.py` evaluate them; `web/prepare_web_data.py` exports precomputed recommendations. A static HTTP server serves `web/index.html` and `data.json`; D3 v7 renders the graph. No account or publication history is required from the student.

Figure 1 shows actual browser captures of the implemented flow. First, selecting a familiar author addresses the publication cold start; the generator chooses up to 50 high-degree seeds with coverage of the eight largest communities. Records are historical UIUC-affiliated authors, with no unverified professor titles or departmental appointments. Numbered communities are inferred collaboration clusters. Second, a D3 force layout supplies the context for recommendations. Solid edges are observed pre-2020 coauthorships; orange dashed edges connect the selected seed to its actual ranked cross-community candidates. Clicking a dashed edge opens that candidate's card. Third, the card places a graph-backed route above topic details, displays autonomy as a positive gauge, and offers an editable outreach template. The template makes no claim that the student works with the seed or knows the intermediary; it does not send messages.

4.2 Shared model and truthful route semantics

The backtest, web exporter, and study generator now import `model.py`. They share the 2019 cutoff, training-only topic profiles, weighted graph, seeded Louvain partition, candidate filter, and Eq. 1. Canonical node/edge order, author-ID tie-breaking, and seeded betweenness remove the earlier sources of run variation. This alignment lets the offline evaluation speak to the displayed ranking, while still not measuring mentoring outcomes.

A bounded shortest-path search supplies only actual two- or three-hop co-authorship chains; when no path is found, the card explicitly says no route was found within three hops and supplies no invented intermediary. This does not imply global disconnection. The student-to-anchor connection is user input; co-authorship does not establish acquaintance or willingness to introduce. Current appointments, availability, and personal trust remain outside the dataset.

The D3 distribution is served locally with its license after verification against the official npm release's SHA-512 integrity metadata. A Chromium walkthrough validates selection, map rendering, dashed-link navigation, card explanations, and the outreach modal without external CDN access. Automated checks compare all 500 displayed recommendations with shared-model rankings, validate every route edge and shortest-path length, and exercise the unavailable-route state on a disconnected graph fixture. The current scaled top-ten bundle contains 483 two-hop and 17 three-hop routes; the unavailable state is tested separately rather than misrepresented as an observed top-ten result.

5 Users and formative feedback

5.1 Population and sampling rationale

Our primary users are UIUC undergraduate and early-stage graduate students seeking research outside their current department. They are a better initial test population than established faculty because cold start and unfamiliar contact routes are central to the problem. Recruitment through ACM student groups, course peers, and interdisciplinary student channels is feasible but introduces convenience-sampling bias.

The authors report five completed individual, 30-minute think-aloud sessions: three CS/CE undergraduate juniors/seniors and two graduate students from Bioengineering and Information Sciences. This mix includes both computing students seeking applications and students in potential destination disciplines. Five sessions are useful for finding formative interaction problems [8]; they are not sufficient to estimate population usability or compare recommendation quality statistically. We do not infer a fixed percentage of problems discovered from the sample size.

5.2 Feedback-to-design trace

The following are author-reported qualitative observations; no participant quotes, issue frequencies, or quantitative usability scores are claimed.

Entry point. Students without publications could not use themselves as a seed. The interface now starts with a searchable author grid and community filters, enabling an instructor to serve as the anchor. A subsequent test should measure whether students can identify a familiar, correctly identified seed without help.

Meaning of dashed links. Participants interpreted dashed lines as weak existing collaborations. The map now explicitly distinguishes observed co-authorship from a missing-tie structural-hole example. The next round should ask participants to explain the legend in their own words; successful navigation alone would not demonstrate comprehension.

Action before explanation. Participants looked for who could introduce them before reading topic details, and a low constraint value appeared to mean a poor candidate. The revised cards place the introduction first and express constraint through an autonomy gauge. The next round should test route comprehension and whether the gauge is mistaken for a success probability.

These changes are visible in the current interface. The subsequent code audit also removed inferred department names and fabricated liaison routes, but neither these corrections nor automated walkthroughs demonstrate that

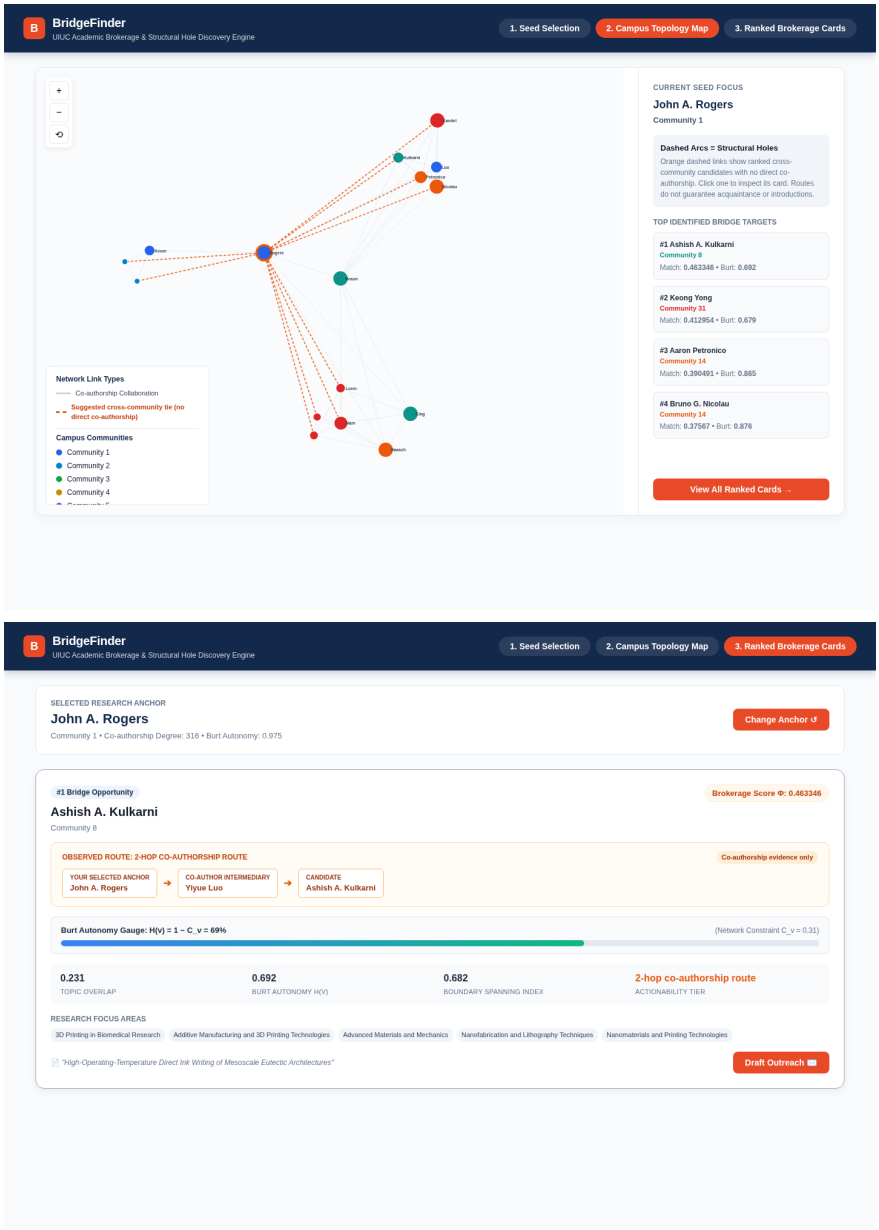


Figure 1: Actual Chromium screenshots: selecting an anchor opens the community map (top); clicking a ranked dashed candidate link opens its recommendation (bottom). The card shows the observed seed–intermediary–candidate co-authorship chain and explicitly limits its claim to co-authorship evidence. Numbered groups avoid presenting inferred communities as verified departments. Full-resolution captures and the automated walkthrough are retained in the repository.

the reported human comprehension problems have been resolved. Anonymized session notes and a repeat human walkthrough would strengthen the audit trail; the repository currently contains neither session-level notes nor completed comparative-study responses.

6 Evaluation: evidence and measurement plan

6.1 Temporal prediction on real data

The backtest predicts new 2020–2024 cross-community co-authorships using only 2015–2019 edges and topic profiles. Seeds must have at least one such future tie with an eligible training node. All methods rank the same non-neighbor, cross-community candidates across retained components. Conditioning on future-positive seeds excludes inactive researchers and makes these results more favorable than

Table 1: Rerun of the shared temporal model: 180 eligible pilot seeds and 1,385 scaled seeds. Full metrics are retained in `midterm/review/`.

	Pilot		Scaled	
Model	P@5	NDCG@10	P@5	NDCG@10
Preferential attach.	0.0011	0.0011	0.0088	0.0133
Betweenness	0.0011	0.0211	0.0078	0.0148
Topic-Only	0.0044	0.0274	0.0208	0.0402
Adamic–Adar	0.0056	0.0297	0.0295	0.0515
Topic+Path	0.0133	0.0615	0.0344	0.0620
Full BridgeFinder	0.0133	0.0514	0.0351	0.0674

an all-student deployment population.

Baselines isolate alternative explanations: preferential attachment tests popularity; sampled global betweenness tests centrality alone; Topic-Only tests topical matching; Adamic–Adar tests common-neighbor evidence. Topic+Path is the simpler ablation. Metrics are macro-averaged over eligible seeds. P@k is the fraction of the top k candidates who become collaborators, reflecting shortlist efficiency. R@10 is the fraction of a seed's eligible future collaborators retrieved, measuring coverage. Binary-relevance NDCG@10 discounts hits by $1/\log_2(r+1)$ and normalizes against the ideal ranking, measuring whether useful names appear early.

Table 1 suggests signal beyond common neighbors, but substantial room for improvement. On the scaled corpus, Full BridgeFinder improves P@5 by 1.19× (0.0351 vs. 0.0295) and NDCG@10 by 0.0159 (0.0674 vs. 0.0515) over Adamic–Adar. Against our proposed targets of 2× P@5 and +0.05 NDCG@10, both scaled targets are missed. On the pilot, the P@5 target is met (2.38×) but the NDCG gain is only 0.0217. These are project targets, not a documented preregistration. Topic+Path ties Full BridgeFinder on pilot P@5 and wins on pilot NDCG@10, while the full model wins both scaled measures. This is mixed evidence for the extra features, not proof that network density causes the difference. Absolute precision remains low: most recommended researchers do not become observed co-authors. Failure to meet an empirical hypothesis is a result to explain, not a reason to change the success criteria after seeing the data.

Reproducibility and uncertainty. The revised shared model canonically orders nodes and edges, uses seed 42 for Louvain and sampled betweenness, and breaks score ties by author ID. A second pilot run under a different Python hash seed produced byte-identical metric JSON. Canonical insertion order changes the partition relative to archived runs: the revised pilot has 180 eligible seeds, and its results must not be interpreted as paired improvements over the earlier 185-seed experiment. We reran both corpora under Python 3.12, NetworkX 3.7, and SciPy 1.18.1; the table reports this implementation, while original archived outputs remain in `prototype/results/`. Dependency pins and test commands are documented in the repository. Uncertainty intervals over historical prediction metrics are still future work. Before tuning weights, retain per-seed metrics, use a separate validation interval, and report paired bootstrap intervals over seeds; overlapping collaboration neighborhoods limit the independence assumption even then.

6.2 Citation impact: a separate observational test

The citation script builds a pre-2020 graph after excluding papers with more than 25 retained UIUC authors. A later paper is cross-community when its known training authors span at least two communities, within-community when exactly one is represented, and unclassified when none is represented. Authors absent from training do not determine its category, so partially observed teams may be misclassified.

The scaled analysis includes 6,231 evaluation papers: 1,071 cross-community, 4,183 within-community, and 977 unclassified. Cross-community mean citations are 64.86 (SD 241.34), versus 61.60 (SD 119.92); medians are 36 and 37. The mean ratio is 1.053, with two-sided Mann–Whitney $U = 2{,}156{,}737$ and $p = 0.0601$. Our success target is a ratio above 1.10 and $p < 0.05$ against within-community papers. Neither criterion is met. Mann–Whitney tests a distributional difference, not specifically a difference in means. Year, field, team size, and repeated authors are uncontrolled, so these results establish neither a causal benefit nor absence of all benefit from brokerage.

6.3 Planned comparative user evaluation

We plan an exploratory within-subject study with 15 students, recruiting three from each of CS, ECE, Bioengineering, Crop Sciences, and Information Sciences. These are recruitment categories, not inferred author appointments. This deliberately broadens the computing-heavy formative sample and tests cross-field comprehension, although it is not representative of all UIUC students. Obtain consent, use pseudonymous identifiers, and allow withdrawal. Participants should first attempt the seed–map–card task, then choose an appropriate research anchor and rate the union of the top five candidates from each method (BridgeFinder, Adamic–Adar, Topic-Only). Identical candidates appear once and are attributed to every method that selected them, preserving five outcomes per method with at most 15 unique cards. All methods use the same cross-community candidate population. Shuffle method-hidden cards independently per participant. The goal is method blinding, not an unverified claim of double blinding.

Relevance and novelty. Seven-point relevance and novelty ratings test whether structural diversity preserves fit while surfacing unfamiliar names. Targets are mean ratings of at least 5/7 for each, compared with both baselines. The primary joint outcome is each participant's fraction of cards rated relevance ≥ 5 and novelty ≥ 5; target a 25-percentage-point gain over Adamic–Adar. Report paired differences and confidence intervals, with a two-sided paired Wilcoxon test at $\alpha = 0.05$ for this single primary comparison; report the mean paired difference and a supplementary

paired-t interval. Treat other comparisons as exploratory and apply Holm correction within their declared family. Do not treat 225 card ratings as 225 independent participants.

Actionability and usability. Seven-point actionability ratings test whether the explanation supports a plausible next step; target at least 5/7 and improvement over Topic-Only. Record unaided task completion and whether participants correctly distinguish observed ties from suggested routes, targeting at least 12 of 15 successful completions and correct explanations. Administer the standard ten-item System Usability Scale after the interactive task; score it using the standard alternating-item transformation, targeting mean SUS $\geq 80/100$. SUS is compared with that fixed target, not a claim that static-card baseline systems have been usability-tested.

Implemented collection and analysis. The browser restricts a session to the selected seed, randomizes unique cards, requires every rating and all ten SUS answers, and exports a UUID-identified session using the analyzer's schema. Missing answers are never replaced with default ratings. The analyzer checks candidate completeness, duplicate identities, valid scales, five outcomes per method, and SUS completeness; invalid sessions are reported as exclusions rather than silently pooled. It scores SUS, tests the primary joint outcome at participant level, and Holm-corrects the exploratory Wilcoxon comparisons. An automated browser fixture completed the export-to-analysis route, and contract tests cover overlap attribution, incomplete data, duplicate IDs, and SUS bounds. These are engineering checks, not human results: generated test answers are discarded, and no comparative human finding is reported. Before recruitment, confirm consent logistics, suitable anchors for each participant, facilitator task/comprehension logging, and private handling of the method key. The small planned sample remains exploratory.

7 Discussion and next steps

BridgeFinder makes network structure part of both candidate selection and explanation. Removing it would remove cross-community filtering, autonomy, and shared-contact explanations, leaving topic search. The present evidence supports a feasible prototype and modest historical ranking gains; it does not establish better mentoring outcomes, usable institutional routes, or a significant citation advantage.

The formative sample is small and self-selected; missing session notes limit independent verification. The graph omits informal relationships, external collaborators, and unpublished work. Harvest caps, large-team cliques, modularity resolution, and heuristic department labels affect who is visible. The shared model and actual route extraction resolve the earlier evaluation/display mismatch and invented institutional paths; current appointments and introduction willingness still require human verification. These limitations guide the next iteration rather than being treated as confirmed benefits.

In early November, select appropriate historical anchors with participants, confirm consent and facilitator procedures, and deploy the validated study workflow. In mid-November, run the 15-person evaluation and report participant-level outcomes, including failures and uncertainty. In late November, revise the interface from observed comprehension errors and tune weights using a held-out validation period. In December, rerun the locked evaluation, document remaining biases, and prepare the final report and demonstration. User feedback and future collaboration precision answer different questions; both are necessary before recommending deployment.

Contributions and AI assistance

Evan Lin: ingestion, backtest pipeline, and campus map. Eric Liu: brokerage score, citation analysis, study tooling, and formative sessions. Both authors contributed to the report. The existing project used Claude for debugging, layout, and typesetting. OpenAI Codex assisted with the rubric audit, shared-model and study-workflow corrections, automated checks, evaluation reruns, screenshots, and editing. No participant responses or quotations were generated for this revision.

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